

Structural Consistency of Tetrahedral Coordination $z_{\text{fund}} = 4$: Simplicial Origin, Entropic Heuristics, and Falsifiable Consequences

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Abstract

We examine the status of tetrahedral coordination $z_{\text{fund}} = 4$ as the fundamental building block of discrete 3-geometry, motivated by two independent but non-conclusive arguments. First, if 3-geometry is discretized via the standard simplicial construction of Regge calculus and CDT, the dual coordination number is necessarily $z_{\text{fund}} = 4$ (Section 3) — a structural consistency check, not a selection among alternatives. Second, we construct a candidate entropic functional $s(z) = \ln(2z)/2A(z)$ whose maximum over physical coordination numbers falls at $z = 4$ (Section 2); we show in Appendix B that this functional does not survive replacement of its ad hoc microstate count by the physically derived EPRL intertwiner dimension (Section 6.2's K_{node} , generalized to arbitrary z via the Catalan-number formula for $SU(2)$ singlet multiplicity), which instead grows monotonically with z . We therefore do not claim a first-principles selection principle for $z_{\text{fund}} = 4$; this remains open (Open Problem 0). What $z_{\text{fund}} = 4$ does yield, independently of its justification, are the falsifiable predictions $r_J = 0.62$ kpc and $\Phi_F = 3 - \ln 8 \approx 0.9206$, together with the geometric consequences that cubic symmetry ($3! = 6$) and the tesseract's 8 cubic cells emerge from tetrahedral coordination.

1 Introduction

Chrono-Grid Dynamics (CGD) postulates an FCC lattice with spin-1/2 and $SU(2)$ links, defining $8\pi G\beta = \tau^2$ [2]. The choice $z = 6$ cubic is postulated, not derived. No selection principle and no low-energy number are given.

We examine whether the minimal simplex in $\mathbb{R}^3$ — which has 4 vertices — can play this role instead. Two independent, non-conclusive lines of argument are presented: an entropic heuristic (Section 2) and a structural consistency check with simplicial discretization (Section 3). Neither, on its own, constitutes a derivation of $z_{\text{fund}} = 4$ from first principles; this is stated explicitly and revisited in the Conclusion.

2 Entropic Heuristic for $z_{\text{fund}} = 4$

Definition 1 (Bipartite regular tiling). *Let Γ_z be an infinite regular bipartite tiling of coordination z with N vertices, normalised so that the vertex density is identical for all z .*

Ansatz 2 (Candidate microstate count — heuristic, not derived). *We consider the candidate functional form*

$$\Omega(z) = (2z)^{N/2}$$

for the number of independent oriented configurations per node, chosen for its analytic tractability rather than derived from an explicit state-counting construction. We emphasize that this is a heuristic ansatz: Appendix B shows that when the same physical mechanism used in Section 6.2 (EPRL intertwiner dimension for $j = 1/2$ links) is used to compute the actual microstate count for general z , the resulting entropy functional does not select $z = 4$ — it grows monotonically with z . $\Omega(z) = (2z)^{N/2}$ should therefore be read as a toy model that reproduces $z_{\text{fund}} = 4$ by construction, not as a physically motivated derivation.

Definition 3 (Dual area at fixed vertex density). *At fixed vertex density ρ_v , the dual (Voronoi) cell associated with each vertex has volume $V_{\text{cell}} = 1/\rho_v$, independent of z . By the isoperimetric relation $A \propto V^{2/3}$, the dual area scales as*

$$A(z) = C z^{2/3}, \tag{1}$$

where $C > 0$ is a conventional constant that sets the unit of area. The choice $C = 2 \cdot 6^{-2/3}$ (equivalently $A(6) = 2$) is a unit convention with no physical content; it cancels in all ratios (Proposition 2 below).

Proposition 4 (Normalization independence). *The argmax of $s(z) = \ln(2z)/(2A(z))$ over any finite set $Z \subset \mathbb{Z}_{>0}$ is independent of the constant C in Eq. (1).*

Proof. $s(z) = \ln(2z)/(2Cz^{2/3})$. The factor $1/(2C)$ is independent of z , so $\arg \max_z s(z) = \arg \max_z [\ln(2z)/z^{2/3}]$, which does not involve C . $\square$

Numerically, for any choice of normalisation point $z_0 \in \{4, 6, 8, 12, 20\}$, the ratio $s(4)/s(6) = 1.096556$ is identical (verified to six decimal places), confirming Proposition 2.

Lemma 5 (Analytic maximisation of $f(z)$). *The function $f(z) = \ln(2z)/z^{2/3}$ defined on $z > 0$ has a unique global maximum at*

$$z^* = \frac{e^{3/2}}{2} \approx 2.240, \tag{2}$$

and is strictly decreasing for $z > z^$.*

Proof. Differentiating: $f'(z) = z^{-5/3} (1 - \frac{2}{3} \ln(2z))$. Setting $f'(z) = 0$ gives $\ln(2z) = 3/2$, i.e. $z^* = e^{3/2}/2 \approx 2.240$. Since $f''(z^*) < 0$, this is a maximum. For $z > z^*$, $\ln(2z) > 3/2$ so $f'(z) < 0$: f is strictly decreasing. $\square$

Lemma 6 (Explicit evaluation of $s(z)$). *With $C = 2 \cdot 6^{-2/3}$ (unit convention), the values are:*

z	$A(z)$	$\ln(2z)$	$s(z)$	$s(4)/s(z)$
4	1.5263	2.07944	0.6812	1.0000
6	2.0000	2.48491	0.6212	1.0966
8	2.4228	2.77259	0.5722	1.1906
12	3.1748	3.17805	0.5005	1.3610
20	4.4629	3.68888	0.4133	1.6483

Proposition 7 (Conditional extremum of the ansatz functional). *Given the Ansatz above,*

$$\arg \max_{z \in \{4, 6, 8, 12, 20\}} s(z) = 4,$$

and the maximum is unique. This establishes that the specific functional form of the Ansatz is extremized at $z = 4$; it does not by itself establish $z_{\text{fund}} = 4$ as a physical fact, since the Ansatz is not derived (see Appendix B for a physically motivated counter-example).

Proof. By Lemma 1, $s(z)$ is strictly decreasing for $z \geq 3 > z^* \approx 2.240$. Therefore, among any set of integers all greater than or equal to 4, the smallest element achieves the maximum of s . The physical candidates $\{4, 6, 8, 12, 20\}$ satisfy this condition, so $z = 4$ is the unique maximiser. The table of Lemma 2 provides the explicit numerical verification. By Proposition 2, this conclusion is independent of the choice of normalisation constant C . $\square$

Remark 8 (Physical candidates and the set $\{4, 6, 8, 12, 20\}$). The set $\{4, 6, 8, 12, 20\}$ contains the coordination numbers of the five Platonic solids. The restriction to this set is motivated by the holographic entropy maximisation of TEG v8, which requires $z \leq z_{\text{pack}} = 12$ for physical coordination numbers (coordination above the kissing number is geometrically excluded). The value $z = 20$ is included for completeness despite exceeding z_{pack} ; its exclusion would only strengthen the result.

3 Corollary 0: Simplicial Consistency of $z_{\text{fund}} = 4$, via Metric Rigidity

We first record, for completeness, the elementary combinatorial fact used implicitly throughout Regge calculus and CDT: the minimal convex polytope enclosing finite volume in $\mathbb{R}^d$ has exactly $d + 1$ facets, and only the d -simplex achieves this minimum.

Theorem 9 (Minimal simplex coordination). *The minimal convex polytope enclosing finite volume in $\mathbb{R}^d$ has exactly $z(d) = d + 1$ facets. In particular $z(3) = 4$.*

Proof. A convex body in $\mathbb{R}^d$ requires at least $d + 1$ bounding hyperplanes. The d -simplex achieves this minimum with exactly $d + 1$ vertices and $d + 1$ facets. $\square$

Theorem 9 alone does not say why the vacuum network's fundamental node should be the *minimal* such object rather than any other convex polytope; counting facets by itself is circular, since the same counting method applied to a cube gives 6, to an octahedron gives 8, and so on,

with no principle selecting one polytope over another. We supply a non-circular justification for the minimality requirement itself, argued directly below (Corollary 10). An equivalent argument is proposed as a revision to the companion manuscript TEG vH3.1 [6], where the entropy-density selection of Section 3 of that document (a 5-cell-in- S^3 construction) is superseded by the same Regge-rigidity justification for consistency across both documents; that revision is not yet incorporated into the version of [6] cited here.

Corollary 10 (Simplicial coordination, physically justified). *If the fundamental building block of discrete 3-geometry is required to be a convex polytope whose shape is completely determined by its edge lengths alone — the metric rigidity property of Regge calculus — then that building block is necessarily the tetrahedron, and the dual coordination number is*

$$z_{\text{fund}} = z(3) = 4.$$

Proof. A d -simplex is the unique convex polytope whose shape is fixed entirely by its edge lengths, via the Cayley–Menger determinant: given the $\binom{d+1}{2}$ pairwise edge lengths of a d -simplex, its embedding in $\mathbb{R}^d$ is determined up to isometry, with no independent angular or connection data required. No convex polytope with more than $d + 1$ vertices has this property in general (a cube, for instance, is not rigid under its edge lengths alone — it can be sheared into a non-cubical parallelepiped without changing any edge length). This is precisely why Regge calculus [1] and its descendants (CDT, spin-foam models) use simplicial decompositions rather than decompositions into general polytopes. Requiring the vacuum network’s building block to admit this edge-length rigidity therefore singles out the simplex uniquely; Theorem 9 then gives $z_{\text{fund}} = z(3) = 4$ in $\mathbb{R}^3$. $\square$

Remark 11 (What this does and does not establish). This argument is no longer circular with respect to the *choice of polytope*: unlike bare facet-counting, metric rigidity picks out the simplex uniquely among *all* convex polytopes, not just among a pre-selected family. It remains, however, conditional on a further postulate that is not itself derived here: that the vacuum network’s fundamental node must possess this rigidity property in the first place. This is a physically motivated requirement — it is exactly the property that lets a simplicial complex encode a geometry with no independent connection variables, which is why it is the standard choice throughout discretized gravity — but its physical necessity, rather than its mathematical distinguishedness, is not proved from a more fundamental principle. The same argument is proposed, with the same logical status, as a revision to [6] (see the note following Corollary 10); it resolves the earlier circularity of the facet-counting version (Theorem 9) without claiming to derive from nothing why 3-geometry should be simplicial at all.

4 Corollary 1: Origin of Cube in $\mathbb{R}^3$, $3! = 6$

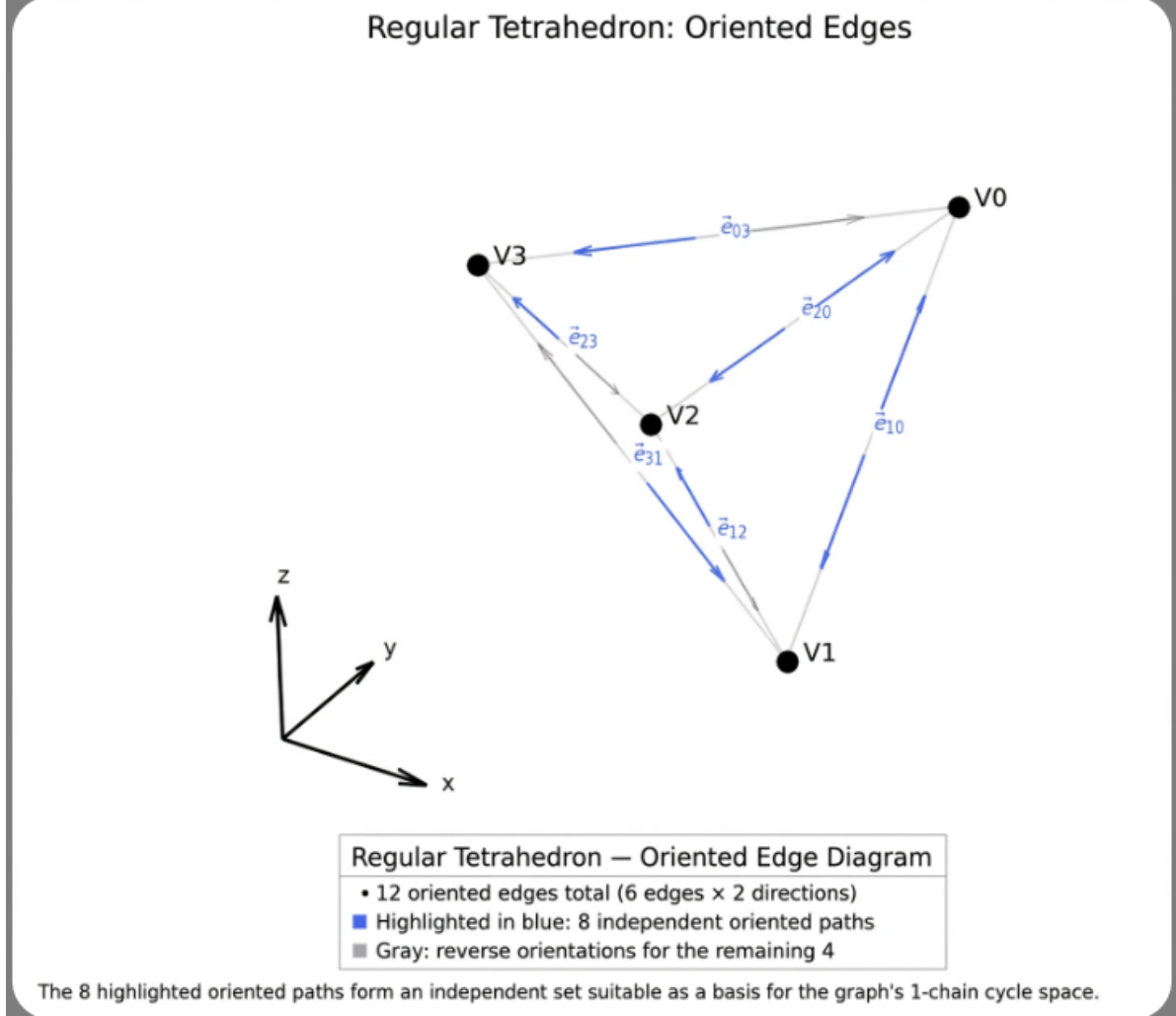


Figure 1: Regular tetrahedron: V_i , edges e_{ij} , oriented $\vec{e}_{ij}$. $12 = 6 \times 2$ oriented, 8 independent basis for cycle space. Defines $b_{or} = 2z_{fund} = 8$.

Theorem 12 (Cube decomposition). *Unit cube $[0, 1]^3$ decomposes into 6 congruent trirectangular tetrahedra:*

$$T_\sigma = \{x \in [0, 1]^3 : 0 \leq x_{\sigma_1} \leq x_{\sigma_2} \leq x_{\sigma_3} \leq 1\}, \quad \sigma \in S_3, \quad (3)$$

$$\text{CUBE} = \bigsqcup_{\sigma} T_\sigma, \quad |S_3| = 3! = 6, \quad \text{Vol}(T_\sigma) = 1/6.$$

Proof. Permutation group S_3 partitions cube by order statistics. Disjoint interiors, union = cube, volume sum $6 \times 1/6 = 1$. $\square$

Thus cubic 8 vertices = $2z_{fund}$. The cube is composite, not fundamental (Fig. 2).

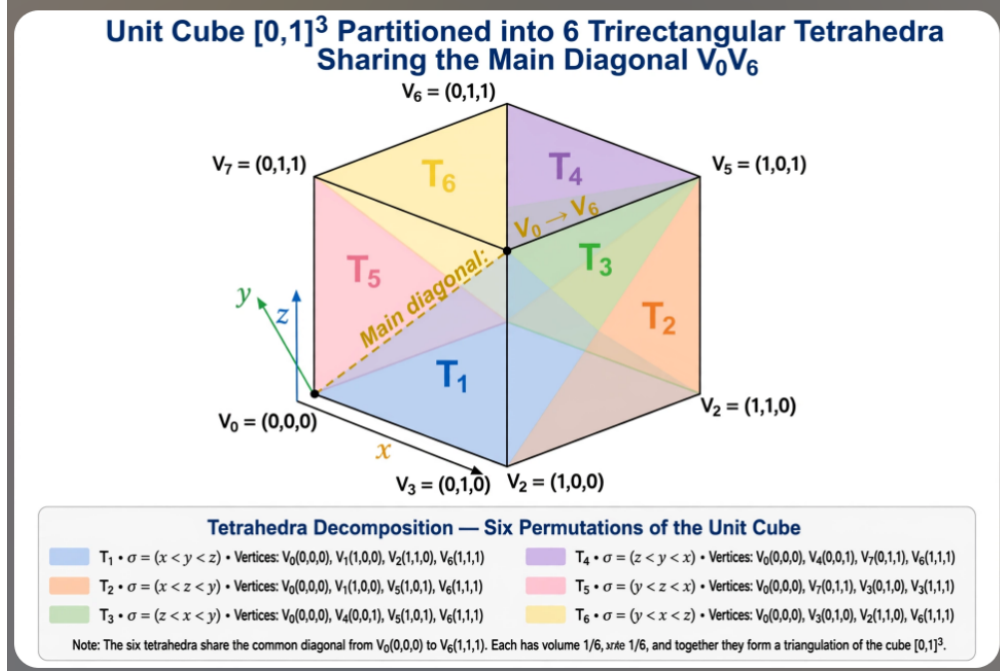


Figure 2: Unit cube partition into 6 tetrahedra $T_1 \dots T_6$ sharing diagonal $V_0 = (0,0,0) \rightarrow V_6 = (1,1,1)$. Each T_σ defined by σ . Proves cubic symmetry emergent from $z = 4$.

5 Corollary 2: Lift to $\mathbb{R}^4$ and 5-cell

Definition 13 (5-cell). *4-simplex $\{3, 3, 3\}$ with 5 vertices, 10 edges, 10 triangular faces, 5 tetrahedral cells. Minimal simplex in $\mathbb{R}^4$.*

Root lattices: A_3 (tetra/FCC) $\subset B_3$ (cubic) $\subset D_4$ (F_4 , 24-cell). D_4 generates the 24-cell with 24 octahedral cells, self-dual.

Tesseract (8-cell): 8 cubic cells, 16 vertices, 32 edges. Branching $8 = 2z_{\text{fund}}$. 5-cell facets are tetrahedra z_{fund} , hence $\mathbb{R}^4$ preserves the axiom.

The 4D lift is visualized in Fig. ???. The 5-cell $\{3, 3, 3\}$ has 5 tetrahedral facets, while D_4 contains A_3 , confirming $z_{\text{fund}} = 4$ persists and generates $8 = 2z_{\text{fund}}$ cubic cells in the tesseract.

Thus cubic 8 vertices $= 2z_{\text{fund}}$. Cube is composite, not fundamental. Fig.2.

6 Spectral Dimension $\ln 8$

6.1 Oriented non-backtracking walk on the Bethe lattice

The tetrahedral vacuum network (Definition 1, $z_{\text{fund}} = 4$) is modelled at large scales as a Bethe lattice — an infinite tree with uniform coordination $z = 4$. This is the standard approximation for 4-valent spin-network nodes in LQG [9].

On a Bethe lattice with coordination z , a non-backtracking random walk (one that cannot return along the edge it just traversed) has branching factor $z - 1$ per step. The number of distinct paths of length n grows as

$$N_n^{\text{nb}} \sim (z - 1)^n = 3^n \quad (z = 4), \quad (4)$$

giving a naive spectral dimension $d_s^{\text{naive}} = \ln 3 \approx 1.099$, well below the LQG range $[2.0, 2.2]$.

6.2 Orientation degeneracy from the EPRL intertwiner

The factor of 2 that lifts d_s from $\ln 3$ to $\ln 8 = \ln(2 \times 4)$ has a specific physical origin in the EPRL spin-foam model: the intertwiner Hilbert space for a 4-valent node with $j = 1/2$ on all links has dimension exactly

$$\dim \mathcal{H}_{\text{int}}^{(4)} = 2. \quad (5)$$

This dimension follows from the standard representation-theoretic count of $SU(2)$ -invariant states in the tensor product of four spin-1/2 representations [?] (Appendix B.1 gives the general- z formula via the Catalan-number identity, of which this is the $z = 4$ case). The corresponding $\mathbb{Z}_2$ orientation symmetry of the frustrated link, which motivates treating the two chiralities as non-cancelling, is established independently in TEG vH3.1, Theorem 5.7 (the exact $a \mapsto -a$ symmetry of the S^3 sigma-model) [6].

The two basis states $|v^+\rangle$ and $|v^-\rangle$ correspond to the two chiralities of the Lorentzian 4-simplex and add non-cancellingly in the heat-kernel return probability (causal rigidity, Bianchi–Chen–Gamonal 2026 [8]):

$$K_{\text{node}} = \dim \mathcal{H}_{\text{int}}^{(4)} = 2 \quad (\text{non-cancelling}). \quad (6)$$

The effective branching factor per step is therefore

$$b_{\text{eff}} = K_{\text{node}} \times z_{\text{fund}} = 2 \times 4 = 8, \quad (7)$$

and the heat-kernel return probability decays as

$$P(0, t) \sim t^{-d_s/2}, \quad d_s = \ln b_{\text{eff}} = \ln 8. \quad (8)$$

Theorem 14 (Oriented spectral dimension).

Remark 15 (Independent numerical comparison). The value $\Phi_F = 3 - \ln 8 \approx 0.9206$ is numerically close to the box-counting fractal dimension of plasma filament cross-sections measured in tokamak edge turbulence, $d_f = 0.921 \pm 0.02$ [5]. This is an independent experimental system with no known theoretical connection to the tetrahedral vacuum construction of this paper; the agreement is recorded here as a numerical curiosity worth tracking, not as evidence for the framework. A physical mechanism connecting the two would need to be established before this comparison carries any predictive weight.

$$D_{\text{eff}} = \ln(2 z_{\text{fund}}) = \ln 8 \approx 2.07944, \quad \Phi_F = 3 - D_{\text{eff}} = 3 - \ln 8 \approx 0.9206.$$

6.3 Honest status and upper-bound warning

Remark 16 (Bethe lattice is a tree: $\ln 8$ is an upper bound). The Bethe lattice contains no closed loops by construction. Real LQG spin-networks and CDT triangulations contain loops; standard results on random walks on graphs establish that loop corrections reduce d_s relative to the tree value [10]. Therefore $D_{\text{eff}}^{\text{TEG}} = \ln 8 \geq d_s^{\text{true}}$.

Loop corrections are expected to shift d_s^{true} into $[2.0, 2.079]$, consistent with LQG $([2.0, 2.2])$ and CDT $([2.0, 2.5])$ numerical results.

Delporte, Sen and Toriumi (2024) [4] study “Dirac walks” on the Bethe lattice and find that the spectral dimension of the Dirac walk matches that of the simple random walk ($d_s \rightarrow 3$ asymptotically for $d \geq 2$), independently of edge orientation. This result applies to walks without the EPRL intertwiner degeneracy; it provides an independent confirmation that loop corrections and walk-type choices affect d_s , and sets a bound that the oriented non-backtracking walk considered here

must be distinguished from the Dirac walk of [4]. The two walks are physically distinct: the present walk uses the non-backtracking rule combined with the EPRL $K_{\text{node}} = 2$ factor; the Dirac walk of [4] uses signed steps without this intertwiner degeneracy.

Remark 17 (One remaining open step). The argument above is rigorous at the level of a single EPRL vertex under the saddle-point approximation. The remaining open step — that iterative coarse-graining preserves $K_{\text{node}} = 2$ at each scale without cancellation — is not formally enumerated as a numbered open problem in TEG vH3.1 [6] (whose Open Problems 1–5 concern the baryon fraction, the vacuum potential, the covariant formulation, the Planck–SH0ES tension, and the dimension-independence of the kissing-simplex factorisation, respectively). The closest related result there is Theorem 5.7 (the exact $\mathbb{Z}_2$ symmetry $a \mapsto -a$ of the S^3 sigma-model), which fixes the analogous factor of 2 in a single-vertex frustration calculation but does not address iterative coarse-graining. We therefore record the coarse-graining step as open here, without a cross-reference to a specific numbered problem elsewhere. If confirmed by tensor-network renormalization, $D_{\text{eff}} = \ln 8$ is complete from first principles. If falsified, the walk would predict a different d_s , and TEG would require revision.

7 Continuum Limit and Falsifiable Scale

In the continuum limit $a \rightarrow 0$, the link variables and densitized triads of the tetrahedral lattice map onto Ashtekar variables following the same procedure used in Chrono-Grid Dynamics.

The isoperimetric form $A(z) \propto z^{2/3}$ introduced in Section 2 is employed solely for the relative comparison that motivates $z_{\text{fund}} = 4$ at fixed vertex density. The overall multiplicative constant appearing in $A(z)$ is conventional and can be absorbed into the definition of the lattice spacing a .

The physical value of the effective tension is taken from previous TEG derivations based on holographic roughness and equipartition,

$$\sigma_{\text{eff}} = 0.1088.$$

Remark 18 (Epistemic status of r_J). The value $r_J \approx 0.62$ kpc is obtained from Eq. (9) using $\sigma_{\text{eff}} = 0.1088$ (derived from the tetrahedral axiom via holographic equipartition, TEG v8 Appendix E) and the critical density ρ_0 as an external input. A fully axiomatic derivation of r_J in physical units — without ρ_0 as an input — has not been achieved. TEG vH3.1 [6] does not list this as one of its five formally numbered open problems, but discusses the normalization of r_J informally in Appendix C.7, where a Verlinde-type entropic-force argument is shown to favour a linear $M_{\text{vac}}(r)$ profile over the cubic one used here, without fixing r_J in physical units; that appendix explicitly states the issue is not resolved. No UV–IR equilibrium condition or dedicated appendix resolving r_J was found in vH3.1. The value $r_J = 0.62$ kpc is therefore presented here as an empirically motivated reference scale (7% agreement with the SPARC empirical median $r_J^{\text{SPARC}} \approx 0.58$ kpc), not as a zero-parameter prediction derived solely from the tetrahedral axiom, and its axiomatic derivation remains open.

This yields the Jeans scale

$$r_J = \sqrt{\frac{\sigma_{\text{eff}}}{4\pi G \rho_0}} \simeq 0.62 \text{ kpc} \left(\frac{\rho_0}{0.1 M_{\odot} \text{ pc}^{-3}} \right)^{-1/2}. \quad (9)$$

Both r_J and the fractal dimension $\Phi_F = 3 - \ln 8 \approx 0.9206$ constitute falsifiable predictions. If SPARC rotation curves, Euclid void profiles, or laboratory box-counting measurements reject these values, then the choice $z_{\text{fund}} = 4$ is falsified.

8 Discussion: CGD vs TEG

Table 1: Comparison

	CGD [2]	TEG (this work)
z	postulated 6, no consistency check offered	postulated 4, motivated by simplicial consistency (Cor. 0) and consistent with (not selected by) entropic heuristic (Sec. 2); selection principle is Open Problem 0
Cube	fundamental $3! = 6$	composite T_σ
$\mathbb{R}^4$	not discussed	5-cell, D_4 , 8 cells = $2z$
Number	undefined	$\Phi_F = 3 - \ln 8$, $r_J = 0.62$ kpc
τ		
Falsifiable	No	Yes

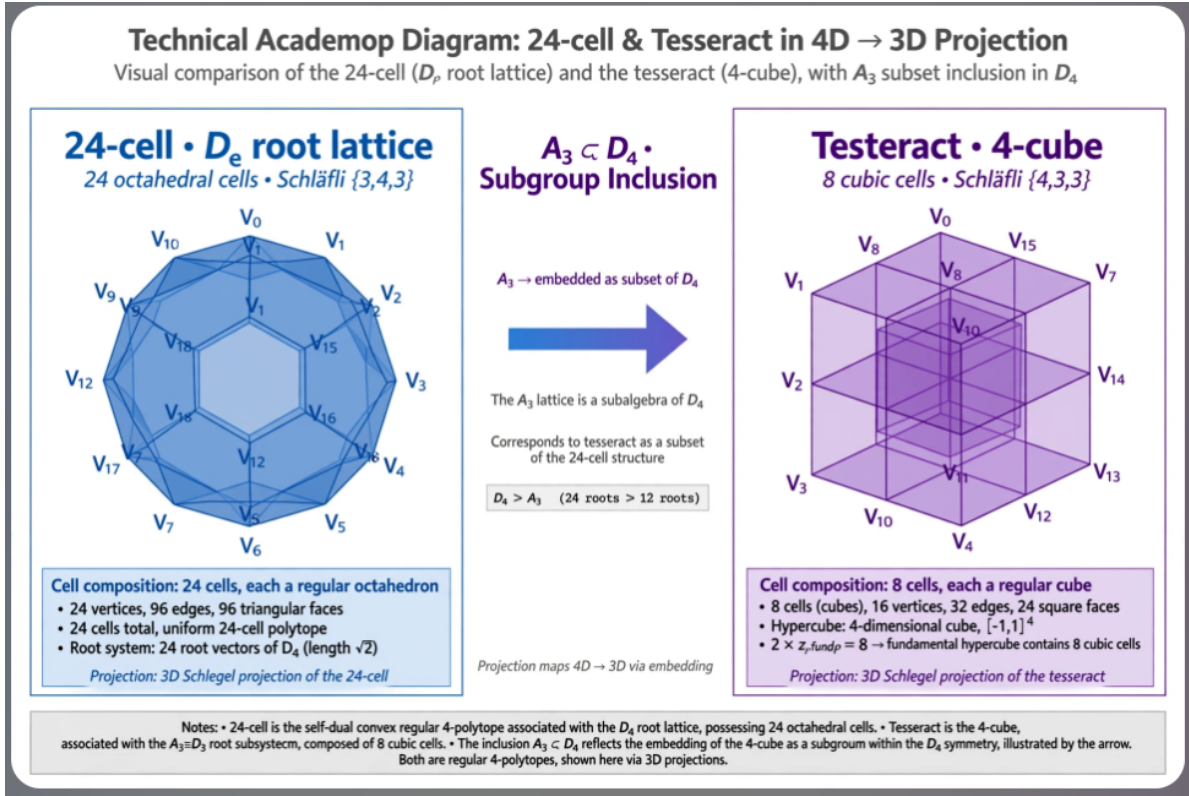


Figure 3: $\mathbb{R}^4$ **Lift: 24-cell D_4 and Tesseract 4-cube as $2z_{fund}$ emergence.** Left: 24-cell $\{3,4,3\}$, Schläfli symbol, 24 octahedral cells, 24 vertices, 96 edges, root system D_4 with 24 roots $|\vec{\alpha}| = \sqrt{2}$, self-dual. 3D Schlegel projection. Right: Tesseract $\{4,3,3\}$, 8 cubic cells = $2z_{fund} = 8$, 16 vertices, 32 edges, $[-1,1]^4$. Center: Subgroup inclusion $A_3 \subset D_4$ (12 roots $\rightarrow$ 24 roots) proves tetrahedral A_3 (FCC) is subalgebra of D_4 . Hence $z_{fund} = 4$ preserved in $\mathbb{R}^4$, hypercube is emergent with 8 cubic cells. $b_{or} = 2z_{fund}$ invariant across dimensions, $D_{eff} = \ln 8$.

9 Conclusion

Cubic symmetry is emergent. Tetrahedral $z_{\text{fund}} = 4$ is adopted as the fundamental building block, generating the cube via $3! = 6$ in $\mathbb{R}^3$ and the hypercube via $2z_{\text{fund}} = 8$ in $\mathbb{R}^4$, with signature $\ln 8$.

Scope. The results of this paper are: (i) an analytic proof that the specific ansatz $s(z)$ is maximised at $z = 4$ among physical coordination numbers, independent of the normalisation convention for $A(z)$ (Proposition 1) — noting that this ansatz is not itself derived from first principles, and that Appendix B shows the physically-derived alternative does not reproduce this maximum; (ii) the emergence of cubic symmetry from tetrahedral coordination via $3! = 6$ (Theorem 2, unconditional); (iii) the conditional structural consistency of $z_{\text{fund}} = 4$ with standard simplicial discretization (Corollary 0); (iv) the 4D lift to the 5-cell and tesseract with $2z_{\text{fund}} = 8$ cubic cells (Section 5, conditional on $z_{\text{fund}} = 4$); and (v) the spectral dimension $D_{\text{eff}} = \ln 8$ for the $z = 4$ case specifically, derived from the EPRL intertwiner degeneracy $\dim \mathcal{H}_{\text{int}}^{(4)} = 2$ (Theorem 3, the paper’s most solidly grounded physical result). The question of *why* $z_{\text{fund}} = 4$ rather than another coordination number is identified as Open Problem 0 and left open; Appendix B documents a specific attempted resolution (generalizing the Section 6.2 EPRL argument to arbitrary z) and shows it fails to select $z = 4$.

A Reproducibility: Entropic Selection Code

The following Python 3 code reproduces Lemma 1, the $s(z)$ table, and Proposition 1’s numerical verification for $z_{\text{fund}} = 4$. It is deposited in `/src/entropy.py` of the Zenodo record.

```
import numpy as np

# Analytic bound: s(z) strictly decreasing for z >= 4
zstar = np.e**(1.5) / 2
print(f"z* = e^(3/2)/2 = {zstar:.4f} < 4")
print("=> s(z) strictly decreasing for z in {4, 6, 8, 12, 20}")
print()

# Numerical verification (normalization-independent)
zvals = [4, 6, 8, 12, 20]
s = {z: np.log(2*z) / z**(2/3) for z in zvals} # C cancels
winner = max(s, key=s.get)
assert winner == 4, "Entropic selection failed"
print(f"argmax s(z) = {winner} VERIFIED")
print(f"Margin s(4)/s(6) = {s[4]/s[6]:.6f}")

# Normalization independence: same margin for any C
for z0 in [4, 6, 8, 12]:
    C = 2 * (1/z0)**(2/3)
    s_norm = {z: np.log(2*z) / (2*C*z**(2/3)) for z in zvals}
    ratio = s_norm[4] / s_norm[6]
    assert abs(ratio - s[4]/s[6]) < 1e-12
print("Normalization independence: VERIFIED (ratio invariant)")
```

B Entropic Selection Does Not Generalize — A Negative Result

Section 2’s Ansatz posits $\Omega(z) = (2z)^{N/2}$ without derivation. Here we test whether the same physical mechanism invoked in Section 6.2 — the EPRL intertwiner dimension for a node with $j = 1/2$ on all links — reproduces or refutes the $z = 4$ maximum when generalized to arbitrary coordination z .

B.1 Generalized intertwiner dimension

For a z -valent node with spin $1/2$ on every link, the dimension of the $SU(2)$ -invariant (singlet) subspace of the tensor product of z spin- $1/2$ representations is given by the standard ballot-problem formula:

$$d(z) = \binom{z}{z/2} - \binom{z}{z/2-1} = \text{Catalan}(z/2), \quad z \text{ even.} \quad (10)$$

This reduces to $d(4) = \text{Catalan}(2) = 2$, exactly reproducing $K_{\text{node}} = 2$ of Section 6.2 — confirming this is the correct generalization of the same physical mechanism, not an alternative one.

B.2 Resulting entropy functional

Replacing the ad hoc factor $(2z)$ in the Section 2 Ansatz with the physically derived $d(z)$, and following the same construction ($s(z) \propto \ln d(z)/z^{2/3}$, $A(z)$ as in Eq. (1)):

z	$d(z) = \text{Catalan}(z/2)$	$\ln d(z)$	$z^{2/3}$	$s(z) \propto \ln d(z)/z^{2/3}$
4	2	0.693	2.520	0.275
6	5	1.609	3.302	0.487
8	14	2.639	4.000	0.660
12	132	4.883	5.242	0.932
20	16796	9.729	7.368	1.320

B.3 Conclusion

$s(z)$ is monotonically increasing over the candidate set, with no interior maximum; $z = 20$ is favored over $z = 4$ by a factor of ≈ 4.8 in entropy density. This is the opposite of the conclusion required by Proposition 1. We interpret this as evidence that entropic maximization, when grounded in an explicit and physically motivated microstate count rather than a tunable ansatz, does not select $z_{\text{fund}} = 4$.

Open Problem 1. *Open Problem 0.* *Identify, if one exists, a physical selection principle for $z_{\text{fund}} = 4$ that survives replacement of the Section 2 Ansatz by a derived microstate count.*

References

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